

Aayush Nandan

aayushnandan623@gmail.com | +91-6299597973 | Portfolio | LinkedIn | GitHub | LeetCode | Codeforces

Professional Summary

Full-stack developer experienced in building web applications with modern JavaScript frameworks and backend APIs. Developed projects involving real-time features and AI-powered components using RAG workflows and vector embeddings. Strong foundation in data structures and algorithms, with a focus on writing clean, reliable, and maintainable code while solving practical, real-world problems.

Technical Skills

Languages: C++, JavaScript, SQL

Backend: Node.js, Express.js, REST APIs, Socket.IO, Apache Kafka, Redis Pub/Sub, JWT

Frontend: React, Next.js, Tailwind CSS

Databases: PostgreSQL, MongoDB (w/ Vector Search), Redis, Prisma ORM

AI/ML: Generative AI, RAG, Embedding Pipelines, Vector Databases, LangChain, Google Gemini

DevOps & Tools: Docker, Git, Postman, AWS (EC2, S3)

Projects

Git Query — Retrieval-Augmented Knowledge Assistant 🔗 October 2025

- Built a Retrieval-Augmented Generation (RAG) system that indexes GitHub repositories into MongoDB (vector embeddings) using Google Gemini, with document chunking for efficient semantic search and contextual Q&A with persistent memory.
- Designed a full-stack application with an AI-driven chat interface to deliver real-time, code-aware insights and semantic understanding for developers.
- Highlights: **MERN, Tailwind CSS, Google Gemini API, Mongo Vector, RAG**

ChatVerse — Scalable Real-Time Chat Application 🔗 June 2025

- Developed a horizontally scalable chat platform using Socket.IO + Redis Pub/Sub for low-latency, real-time messaging across distributed Node.js servers.
- Integrated Apache Kafka for high-throughput, reliable batch message processing, with a dedicated Kafka consumer persisting chat streams into PostgreSQL via Prisma ORM for durability and efficient retrieval.
- Highlights: **Node.js, Socket.IO, Redis Pub/Sub, Apache Kafka, PostgreSQL, Prisma, Next.js**

Digital Banking Application 🔗 December 2024

- Implemented ACID-compliant transactions using PostgreSQL, ensuring data consistency and reliability across critical operations such as fund transfers, deposits, and withdrawals.
- Secured the application with JWT-based authentication and enforced Zod validation across APIs to guarantee type safety, input correctness, and resilience against malformed requests.
- Highlights: **MERN, PostgreSQL, Zod Validation, JWT, Bcrypt**

Education

Chandigarh University, BE in Computer Science and Engineering August 2022 – June 2026

- CGPA: 8.58

St. Michael's High School, Senior Secondary , CBSE 2019 – 2021

- Percentage: 94.8

The Tribhuvan School, Secondary , CBSE 2017 – 2019

- Percentage: 96.2

Achievements

- Solved **450+** questions on LeetCode and secured **Global Rank 703** in LeetCode Weekly Contest 475.
- Achieved the **top 25** position out of 1000+ applicants in the Hack with Chandigarh Hackathon.
- Achieved the **top 10** position out of 150+ teams in the Get Set Pitch Ideathon.